ECE 5360

Q1

a)

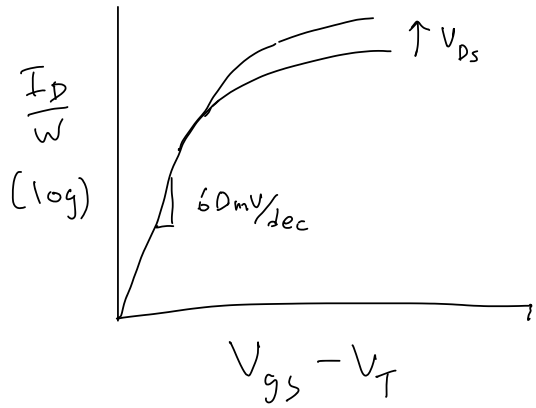
	$V_{GS} > V_{Th}$ On	$(V_{GS} < V_{Th})$ off
Ballistic 2D	$\frac{I_D}{W} \propto V_{gs}^{3/2}$ No dependence on V_{DS}	Exponential Dependence. $\frac{I_D}{W} \propto e^{V_{gs}}$
Long Channel	$\frac{I_D}{W} \propto V_{gs}^{2}$ No dependence on V_{DS} if $\lambda \to 0$. otherwise $\propto (1 + \lambda V_{DS})$	Same exponential dependence. $\frac{I_D}{W} \propto e^{V_{gs}}$

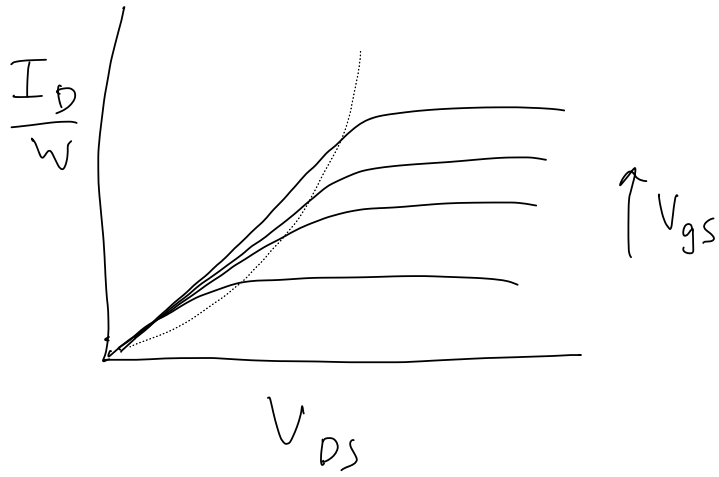
b)

There are several tradeoffs involved to achieve the highest on current. For example, shorter channels increase ballistic transport but reduce reliability and increase current leakage. High mobility materials like GaN enhance performance but complicate fabrication among other tradeoffs. From the discussion, another tradeoff is that when you have a large effective mass m*, you should pair it with a lower q_v material, but for a small effective mass, you should design it with a higher q_v. This is because it's proportional to 1/(m*sqrt(q_v)). For a higher on current, you can also have a very small mass m_c but this generally means a small band gap, which can lead to tunneling. These are just a couple of the discussed tradeoffs.

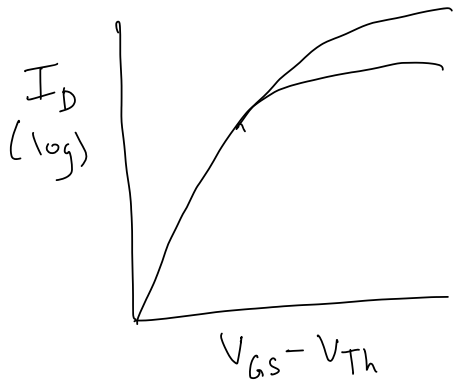
The impact of **contact resistance** is that it will reduce the on current for a given voltage. Actual on current will be lower. This impacts tradeoffs, as it sets a bound on performance that is worse than the quantum expected limit. To get the best on current, you need to consider this when choosing design material and parameters. If sacrifices are made to yield the best on current, you also need to consider whether those tradeoffs also impact contact resistance. For example, switching to a 10FET (tenfold) for on current performance may be a poor choice do to increased contact resistance.

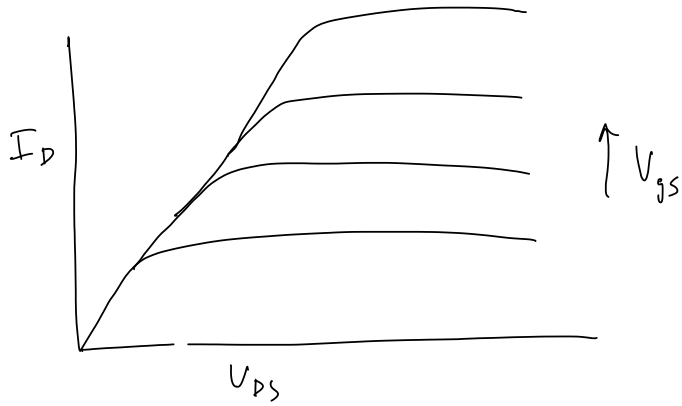
c) 2D FET Ballistic





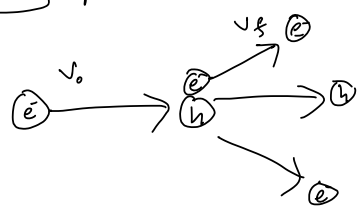
1D FET Ballistic





Similarities include that they have the same general shape and are characterized by ballistic transport. No dependence on L since there is no scattering. We also see both have a high mobility. The graph shapes are also similar. It follows a cutoff to saturation to linear regime as you change Vgs. As for differences, they have slightly different equation dependencies. 2D current is proportional to Vgs^(3/2), but in 1D, that dependency is a Vgs^2. For the IV curve, the 1D plot follows the line of quantized conductance in the linear regime (looks like a resistor), whereas in 2D, it moreso follows a curve. 1D also exhibits far fewer modes, while 2D has many modes/states. 1D also has a higher current flow to charge ratio, as there is less freedom for current direction than 2D.

Q2) MKC 4.11



By symmetry, all particles move at v_f after collision.

Energy: $\frac{1}{2}mv_0^2 = \frac{3}{2}mv_f^2$ Momentum: $mv_0 = 3mv_f$

Recall, $E_g = \Delta PE \Rightarrow E_0 = \frac{1}{2}mv_0^2 = E_g + \frac{3}{2}mv_f^2$

Note by momentum, $v_f = \frac{v_0}{3} \Rightarrow E_0 = E_g + \frac{3}{2}m\frac{v_0^2}{9} = E_g + E_0/3$

$\Rightarrow \frac{2}{3}E_0 = E_g \Rightarrow \boxed{E_0 = \frac{3}{2}E_g}$ As desired

Q3) MKC 4.13

From 4.4.20, $I = q_h A N v \theta$

From background, $A = 10^{-5}\,cm^2$, $N = 10^{22}\,cm^{-3}$, $v = 10^7\,cm/s$, $I = 10mA$

$E_g = 1.1eV$

Using conductivity effective mass, $B = \frac{4\sqrt{2m^*}\,E_g^{3/2}}{3q_h\hbar} = \sqrt{0.25}\cdot 7.87\cdot 10^7 = 4\cdot 10^7$

Recall $\theta = \exp\left(-\frac{q_h B L}{E_g}\right)$

$\theta = \frac{I}{q_h A N v} = 6.2\cdot 10^{-8} = \exp\left(-\frac{q_h B L}{E_g}\right) \Rightarrow \boxed{L = 4.5nm}$

The field is $\mathcal{E} = \frac{E_g}{q_h L} = \boxed{\mathcal{E} = 2.4\cdot 10^6\ \frac{V}{cm}}$

$\Rightarrow$ Same order of magnitude (10^6, nm scale), so matches expected in textbook ✓

Q4) MKC 10.14

Following section 10.5 example $\frac{Q_{fg} d_1}{\epsilon_{ox}} = 1V$ $\epsilon_{ox} = 3.9\epsilon_0$

By diagram $d_1 = d_2 = x_{ox} = 10nm \Rightarrow Q_{fg} = 3.453 \cdot 10^{-7} \frac{C}{cm^2}$

$\Rightarrow$ inject $Q_{fg} \cdot 0.5\mu m \cdot 100\mu m = 1.73fC$

$V_{GF} = 10 \frac{C_{ox}}{C_{ox} + (2/3)C_{ox}} \Rightarrow 5V$ due to V_T off by 1V

We now follow the $\mathcal{E}_{eff} \rightarrow \mu_{eff} \rightarrow V_{Dsat} \rightarrow \mathcal{E}_{sat} \rightarrow I_{Dsat}$ chain

This process was described in example and last homework.

I have a python script to compute this now automatically.

$\Rightarrow I_{Dsat} = 62mA$ for $100\mu m \times 0.5\mu m$ channel/FG.

$\Rightarrow I_G = \underbrace{C}_{10^{-3}} I_{Dsat} \exp\left(-\frac{\phi_B}{q\mathcal{E}_m \lambda}\right) \approx 3.7fA$

↳ computed as $2.4 \cdot 10^5 \frac{V}{cm}$ by python.

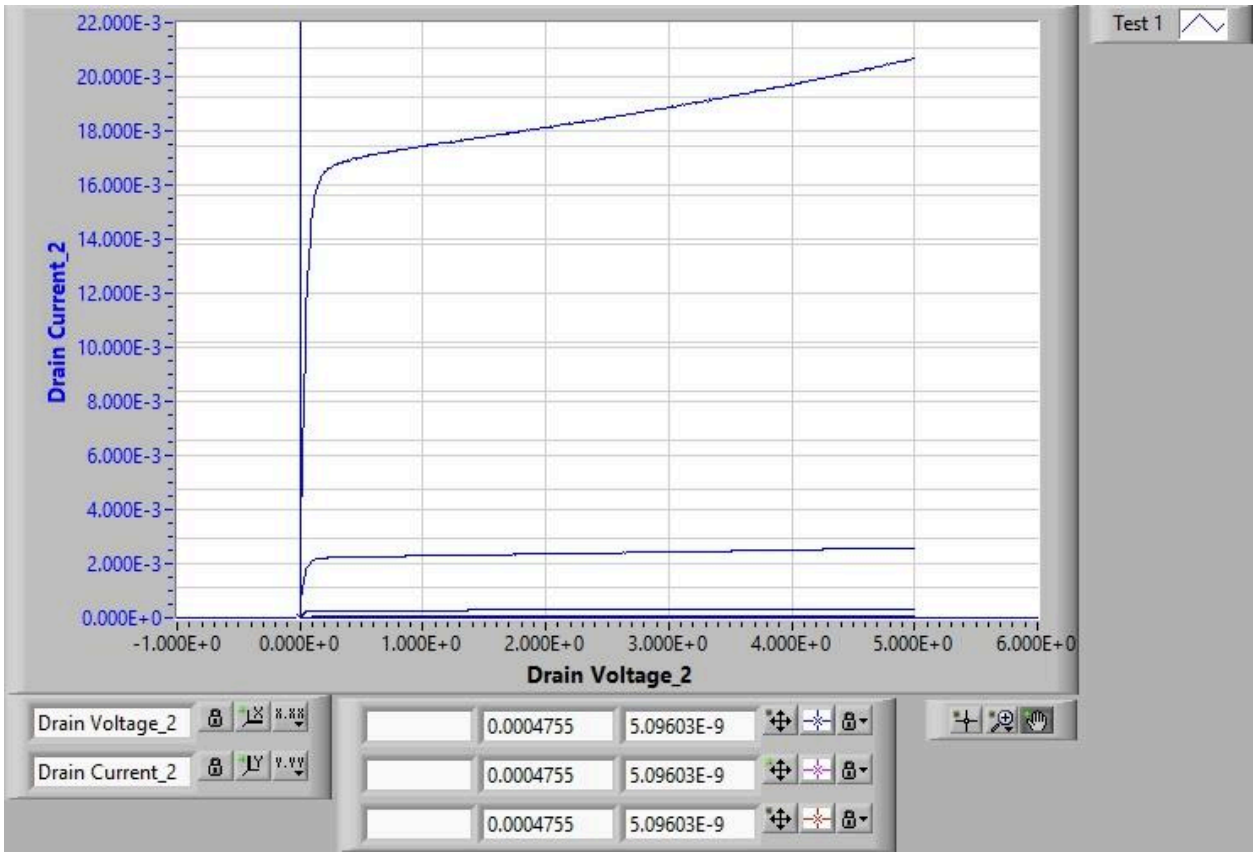
$\Rightarrow \bar{I} = \frac{3.7 + (\text{basically zero})}{2} = 1.85$

$t = \frac{1.73fC}{1.85fA} = 0.935s \approx \boxed{t = 0.9s}$ ✓ Matches appendix answer

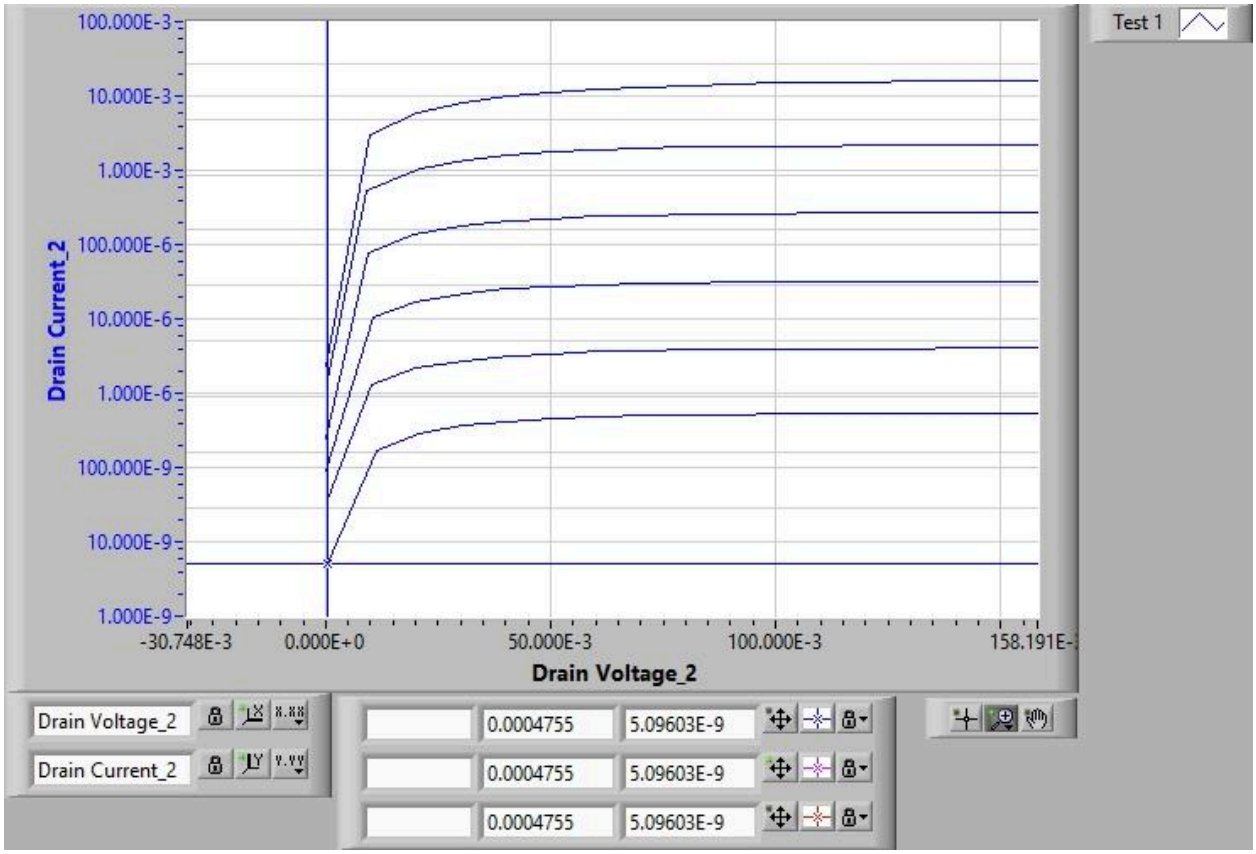
Lab 8

(a) Plot the transfer and output characteristics of the GaN HEMT and compare and comment on the similarities and differences with the Silicon MOSFETs.

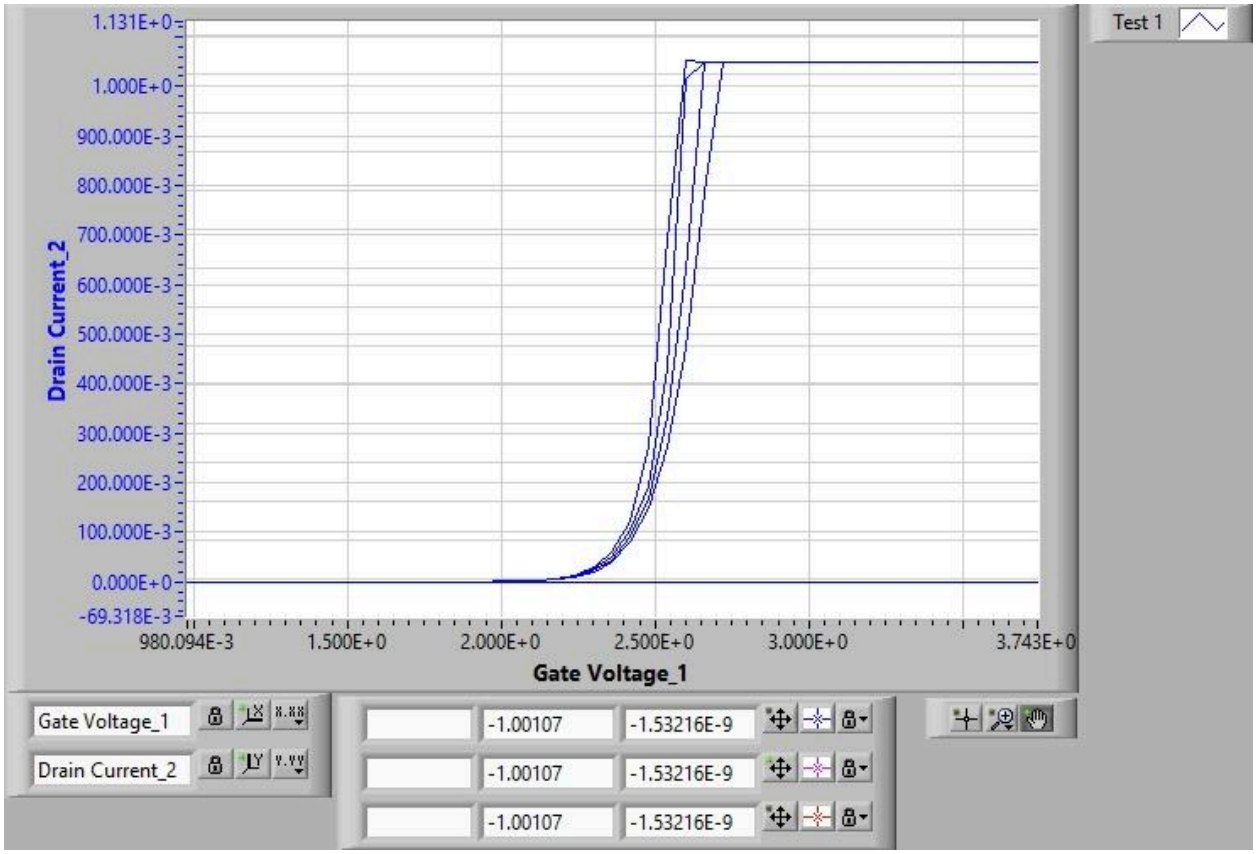
The desired plots are shown below:



Drain Current vs Drain Voltage
(Step Gate V)



Drain Current vs Drain Voltage
(Step Gate V, semilog)



Drain Current vs Gate Voltage
(Step Drain V)

Compared to our previous lab results, GaN HEMTs show a steeper transfer curve with higher transconductance compared to Silicon MOSFETs. Furthermore, GaN HEMTs seem to be more linear at low drain voltages and have a sharper/less smooth saturation, reflecting their superior electron mobility and lower on-resistance. Silicon MOSFETs by contrast have smoother transitions between regions. These distinctions highlight GaN HEMTs' advantages in high-frequency and high-power applications, whereas Silicon MOSFETs remain more suited for cost-sensitive, lower-power circuits.

(b) From the spec sheet, you will notice that the device actually has a GaN HEMT connected to a Silicon transistor. What is the reason it is in this form?

The GaN HEMT is paired with a Silicon transistor because the GaN can handle high voltages efficiently due to its higher breakdown field, protecting the silicon device while leveraging the silicon device's control and processing advantages. The GaN HEMT shields the silicon transistor so that it doesn't experience any sufficiently high voltage that could damage the silicon transistor's integrity.

(c) Read up on the new applications enabled by GaN HEMTs and write a 1-paragraph summary.

From my readings, GaN HEMTs are extremely useful in high-frequency and high-power systems. This is due to their superior electron mobility and wide bandgap. This means that their use has enabled 5G base stations, high-power electric vehicle charges, data center power supplies, and renewable energy systems, where efficiency and compactness are critical. Their fast switching speeds and reduced losses also make them ideal for RF amplifiers in satellites, radar systems, and wireless communications. In these applications, GaN HEMTs are superior to traditional silicon devices and drive innovation in modern high-frequency devices and power electronics.